

G D HRUTHIK

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 3+ years of experience designing and deploying end-to-end ML pipelines and production-grade models. Skilled in Python, TensorFlow, and PyTorch, with a strong record in computer vision and anomaly detection projects. Proven success in driving measurable improvements across healthcare and enterprise domains. Eager to contribute ML expertise to GenAI innovations and data security advancements.

EDUCATION

New England College, Henniker, NH
Master of Science, Data Science and Analytics
• **GPA:** 3.74

Jun 2025

PROFESSIONAL EXPERIENCE

Sanofi

Apr 2024 - Present

Machine Learning Engineer

Boston, MA

- Collaborated with clinical teams to set up ML pipelines for genomic and clinical trial data using clean Python code, enhancing experiment reproducibility.
- Containerized models with Docker and Kubernetes to enable scalable deployment across oncology and immunology studies.
- Implemented model monitoring, versioning, and compliance checks in alignment with FDA and GxP requirements.
- Developed and documented internal AI tools with Streamlit and FastAPI, boosting adoption among scientific teams by 40%.
- Integrated EHR and biomarker data into production pipelines, supporting translational medicine and biomarker discovery.
- Engineered a computer vision workflow with CNNs and PyTorch for microscopy images, reducing protein misfolding detection time by 50%.

Cognizant Technology Services

Aug 2021 - May 2023

Machine Learning Engineer

Coimbatore, TN, India

- Designed real-time anomaly detection models with Autoencoders and Isolation Forests in Python, reducing equipment downtime by 20% for industrial clients.
- Built predictive maintenance pipelines using Random Forest, Kafka, and Azure Stream Analytics to forecast equipment failures.
- Developed and documented a fraud detection solution in Azure ML that achieved inference speeds under 120ms and improved detection accuracy.
- Supported an e-commerce recommendation system in TensorFlow, leading to a 15% increase in click-through rates during A/B testing.
- Collaborated on end-to-end ML operations pipelines, standardizing containerization and monitoring processes, and contributing to team discussions for process optimization.

TECHNICAL SKILLS

- **Programming & Data Analysis:** Python, C++, R, SQL, Scala, Pandas, NumPy
- **Machine Learning:** Regression, Classification, Clustering, Dimensionality Reduction, Ensemble Methods (Bagging, Boosting, Stacking), Hyperparameter Tuning (Optuna, Grid Search)
- **Deep Learning & Computer Vision:** CNN, RNN, LSTM, BERT, CodonBert, ProGen2 Transformers, Attention Mechanisms, Reinforcement Learning, CLIP, Image Classification, Object Detection, Fine-tuning Models, Diffusion Models
- **Frameworks & Libraries:** PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, LightGBM, DeepSpeed, Flask, Django
- **Natural Language Processing:** Text Preprocessing, Tokenization, Word Embeddings, Sentiment Analysis, LangChain, RAG, Hugging Face Transformers, OpenAI API
- **Generative AI:** GPT-based LLMs, Diffusion Models, Prompt Engineering
- **Vector Search & Retrieval:** FAISS, Pinecone, Chroma, Weaviate
- **MLOps & Model Deployment:** MLflow, DVC, Docker, Kubernetes, Azure Kubernetes Service (AKS), Streamlit, Ray, TorchServe, FastAPI, Triton Inference Server, TensorFlow Serving
- **CI/CD & Workflow Orchestration:** Jenkins, GitHub Actions, Apache Airflow, Azure DevOps, Luigi
- **Cloud Platforms:** AWS SageMaker, GCP Vertex AI, Azure ML Studio
- **Visualization & Reporting:** Tableau, Matplotlib, Seaborn
- **Data Engineering:** Hadoop, PySpark, Azure Databricks, Data Factory, Synapse Analytics, Snowflake
- **Project & Collaboration Tools:** Jupyter, Google Colab, VS Code, Agile, A/B Testing, Requirement Gathering